

SMARTATTEND

Google Sheets Stack - University Deployment Strategy & Costs

This document outlines the operational roadmap, cost estimates, and timeline required to scale and deploy the current Google Sheets + Google Apps Script + Netlify PWA system across a university campus.

1. TECHNICAL DEPLOYMENT STEPS

STEP 1: UPGRADE TO GOOGLE WORKSPACE (CRITICAL FOR EMAILS)

Current State: Personal @gmail.com accounts are capped at sending a maximum of 100 emails per day. In a standard class size, this limit will block the parent and student notifications immediately.

Solution: Upgrade to a Google Workspace Business account (Starter or Standard). This increases the Google Apps Script email sending limit (MailApp/GmailApp) to 1,500 emails per day.

STEP 2: CONNECT A CUSTOM DOMAIN

Current State: The app is deployed under a random Netlify subdomain.

Solution: Map a professional domain or university subdomain (for example, smartattend.yourcollege.edu) to Vercel or Netlify to make it institutional.

STEP 3: SETUP ACADEMIC SHEETS & BULK IMPORT

Current State: Hand-entered mock data lists.

Solution: Standardize spreadsheet formats for departments and classes. Create CSV/Excel templates allowing administrators to upload entire student registries and teacher timetables in bulk to Google Sheets.

STEP 4: CLASSROOM HARDWARE SETUP (OPTIONAL)

Current State: Relies on student/teacher personal devices.

Solution: Mount budget tablets or static cameras at classroom entrance doors assigned with fixed IP addresses to match the IP Management System validation.

2. COST ESTIMATION & BUDGET BREAKDOWN

Because the application leverages Google's free serverless infrastructure (Sheets and Apps Script) and free static web hosting, the ongoing running costs are extremely low.

Expense Item	Billing Type	Estimated Cost	Description
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Google Workspace Account	Monthly	\$6.00 - \$12.00 / mo	Required for 1,500 daily email capacity. Only 1 administrator account is needed.
Custom Domain Name	Yearly	\$10.00 - \$15.00 / yr	Institutional branding (e.g., .edu or .com domain).
Netlify/Vercel Hosting	Monthly	\$0.00 (Free Tier)	Free hosting plan easily handles static web assets for high traffic.
Classroom Tablets & Mounts	One-time	\$100.00 - \$150.00 / rm	Optional hardware mounted at classroom doorways.

TOTAL SOFTWARE OPERATIONAL COST: ~\$10.00 to \$15.00 per month.

3. DEPLOYMENT TIMELINE & ROLLOUT SCHEDULE (2-3 WEEKS)

WEEK 1: INITIAL SYSTEM SETUP & DOMAIN ROUTING (3 - 5 DAYS)

- Register and configure the new Google Workspace administrator account.
- Migrate all Apps Script code to the Workspace account and redeploy the API.
- Purchase and map the custom domain to the Netlify platform.
- Format and upload all course databases, student directories, and teacher schedules.

WEEK 2: DEPARTMENTAL PILOT RUN (7 DAYS)

- Deploy and test the live application in a single class (30-50 students).
- Monitor camera scanning angles, classroom lighting variations, and check cellular connectivity speeds within target campus locations.

WEEK 3: GENERAL CAMPUS ROLLOUT (2 - 3 DAYS)

- Distribute the PWA/APK download links to all students in bulk.
- Brief and onboard the remaining department professors.
- Formally initiate live attendance logging and automated parent email alerts.